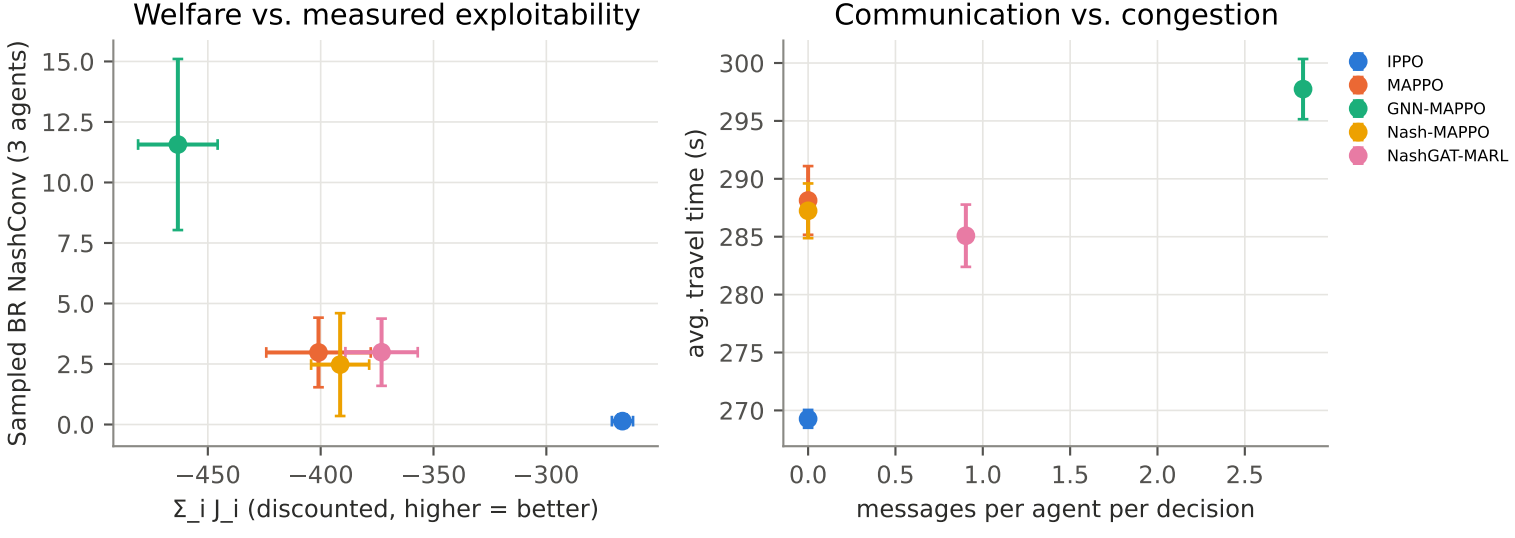
Welfare vs. measured exploitability
Sampled BR NashConv (3 agents)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−450
−400
−350
−300
Σ_i J_i (discounted, higher = better)
Communication vs. congestion
avg. travel time (s)
300
295
290
285
280
275
270
0.0
0.5
1.0
1.5
2.0
2.5
messages per agent per decision
IPPO
MAPPO
GNN-MAPPO
Nash-MAPPO
NashGAT-MARL